

Designing an Inclinometer Stand



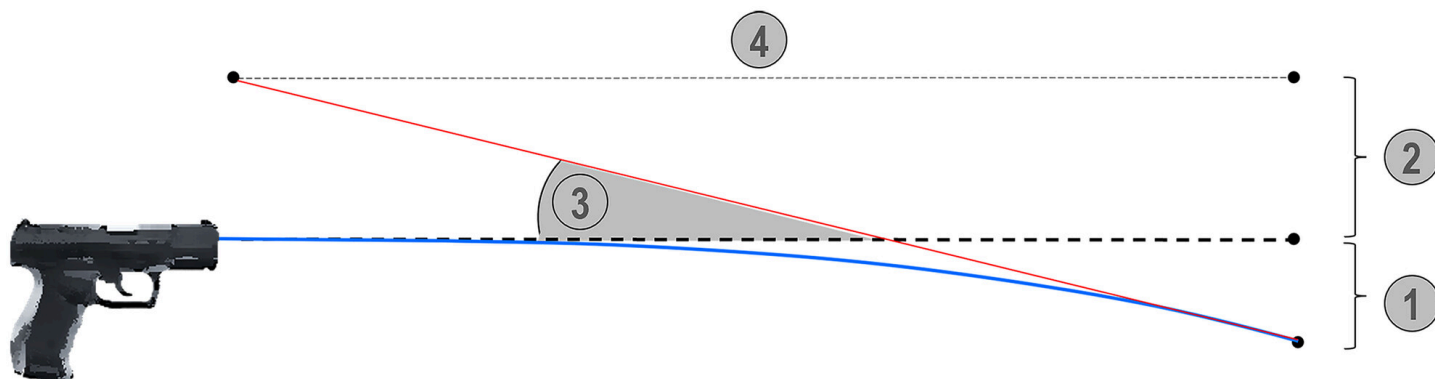
A1—Problem Identification

Dr. Kovacs' problem is that when he wants to teach his forensic students, he needs to hold the inclinometer in his hand; otherwise, the most accurate result will not be shown. Due to "old age" and "caffeine addiction", he can't hold an inclinometer perfectly still since his hand will tremble, which leads to an inaccurate measurement of the trajectory of the bullet, so he can't give his student a clear exposition of concepts he wishes to demonstrate in forensics, such as bullet trajectory in a crime scene. An inclinometer is a tool used in Dr. Kovacs' forensics class that can measure the angle between a line formed by two points and the ground. The side with a screen that indicates the angle measurement is the front side. It has two lasers on its left and right sides, so the two points will be opposite of each other. It can only find the angle if its two lasers point to some surface. It can't find the angle when it's lying down (in other words, the front side is pointing to the roof or the floor). He doesn't have a proper way to hold the inclinometer still after adjustment, so our goal is to build such an adjustable supporter that can hold the inclinometer still after adjustment for the inclinometer with a side length of 2.5".

A2—Problem Definition

Dr. Kovacs and his forensics students will use the inclinometer during class and lab work, where it has to be held manually. In Dr. K's class, he will use it for teaching and demonstrations, and his students (31 per class on average) will use it for labs and practicing exercises. Every school year, they need to use it for 2 weeks (excluding weekends). During these days, each of his students will use the inclinometer twice per day, where a total of 620 measurements will be taken within these 2 weeks. Since there's no dedicated support that can hold the inclinometer securely, these class activities will cost more time on double-checking the measurement, and will increase the probability of inaccurate exposition, and even the incidence of dropping the inclinometer, which may add extra cost on a broken device. Some of the wilder students in Dr. K's class may play with the device, such as pointing the laser emitter at another student's eyes, which may cause safety issues. Dr. Kovacs doesn't want any of these things to happen, which is why he needs our support for his inclinometers in forensics class.

A3—Problem Elaboration



① Bullet drop (BD)

② Vertical offset (VO)

③ Drop angle (DA)

④ Given distance (d)

----- Line of departure (LoD)

——— Bullet Flight Path (BFP)

In forensic crime scene evaluation, Mattijssen and Kerkhoff explain that bullet trajectories are estimated from measurements and that the accuracy and precision of those measurements vary depending on the method used, which shows that unstable or inconsistent angle setup is not only a classroom issue but a broader measurement problem (*ScienceDirect*). A similar concern appears outside forensics: in avalanche safety, McCammon reports that inclinometers are key tools for measuring slope steepness, yet even under ideal conditions currently available, inclinometers have an accuracy of about $\pm 4^\circ$, and that uncertainty can increase unintended exposure to avalanche risk (*Montana State University*). In landslide monitoring, the U.S. Geological Survey uses both inclinometers and lasers to track slope movement because errors or missed changes can affect buildings, transportation routes, and human safety (*USGS*). This means solving it could benefit future forensic practitioners, investigators, and other professionals who rely on reliable positionings and repeated measurements.

A4—Concern of Primary Stakeholders

One of our primary stakeholders is Dr. Kovacs because he needs to use this inclinometer to teach his students each year in forensic class, and doesn't want the teaching process delayed due to inaccurate measurements. Another primary stakeholder is Dr. K's forensic students, since they must use the inclinometer for class activities. Additionally, especially for lab activities, if every student gets a different measurement for the same trajectory, the lab result may not be reliable, thereby causing a grading crisis for both Dr. Kovacs and his students. Also because the hand-holding method for the inclinometer is not stable and could result in it dropping to the ground, this will damage the inclinometer, and either Dr. K needs to buy a new one or the student who breaks the inclinometer needs to buy a new one, which creates unnecessary cost and delays the finishing time of the unit that requires inclinometers. The other primary stakeholder is other teachers who also teach forensics because they will teach the same as Dr. K does, and thereby they need to use the inclinometer. The secondary stakeholders could be civil engineering students, because inclinometers can also be used to monitor the tilt of bridges, high-rise buildings, and dams, to ensure stability by providing precise measurements of movement, tilt and deformation which allows engineers to detect structural risks before a disaster happens, and also people who would want to find the height of something too tall to access. After all, it could be used to find the angle of elevation, which could then be used with the distance from the base of the object to find the height.

A5 & A6—Problem Justification

Forensic work depends on angle-based measurements that must be accurate and consistent, not just approximate. In Dr. Kovacs's class alone, the inclinometer may be used about 620 times in two weeks because about 31 students use it twice per day over the unit, so even a small measurement inconsistency can affect many trials. McCammon reports that inclinometers can have an accuracy of about $\pm 4^\circ$ even under ideal conditions, which shows that angle measurement already contains uncertainty before hand tremor and student-to-student variation are added (*Montana State University*). In addition, Mattijssen and Kerkhoff explain that bullet trajectory results vary depending on the measurement method (*ScienceDirect*), and Poulton's WVU research also shows that some of the largest trajectory errors were affected by measurement method and participant skill (*The Research Repository@WVU*). Together, these findings support the idea that repeated hand-held use can introduce too much operator variation. Because Dr. Kovacs and his students currently have no dedicated stand, the system depends on human steadiness for each of those 620 measurements, which increases the chance of inconsistent readings, extra rechecks, and wasted class time.

Works cited

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